# Md Tanzeel Adam Khan

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## Education

**Indian Institute of Technology Patna** — 2024 – 2026
*Master of Technology in Artificial Intelligence (CGPA: 8.5/10)* — *Patna, India*

**Haldia Institute of Technology** — 2020 – 2024
*Bachelor of Technology in CSE – AI & ML (CGPA: 8.9/10)* — *Haldia, India*

## Publication

**When Should Language Models Search? A Topology-Aware Analysis** | *Under Review, TMLR*

- First Author | Proposed A* search that explores multiple reasoning paths an LLM can take, with formally proven admissible heuristic ($h(s) \leq h^*(s)$). Evaluated across **8 benchmarks** $\times$ **3 scales (0.5B–14B)** $\times$ **4 methods** = 96 configurations, 500 instances each.
- Designed a zero-cost Topology Router (no LLM inference needed) using 14 structural features; **closes 40% of the gap** between the best single method and a perfect oracle. **95% routing accuracy** on CodeContests. Key feature predicts method success at $R^2 = 0.94$.

## Research Experience

**Indian Institute of Technology Patna** — Jul 2024 – Jun 2026
*M.Tech Thesis | Advisor: Prof. Asif Ekbal | LLM Reasoning & Search* — *Patna, India*

- Implemented A* search with true branching ($K$=3 parallel LLM calls — solving an API limitation where only 1 response is returned), self-consistency voting over multiple goal nodes, entity-coverage heuristic (guides search toward unaddressed question aspects), and repetitive output filtering. Ran **48,000+** paired evaluations with Wilson confidence intervals and McNemar significance tests.
- Engineered a routing pipeline that predicts the best reasoning strategy from problem structure alone: 13 structural features, multi-classifier optimization (XGBoost/GradientBoosting, 5-fold CV, 100 seeds), and analysis of when each method wins/fails. Discovered Fluency-Correctness Asymmetry: **38%** of fluent traces are incorrect — explaining why probability-based routers fail.

## Professional Experience

**Mercedes-Benz Research & Development India (MBRDI)** — Nov 2025 – Aug 2026
*AI Engineer Intern | AMG System Software Validation Team* — *Bengaluru, India*

- **iFAST (Intelligent Failure Analysis through Signal Tracing):** Architected an autonomous AI agent (Claude/AWS Bedrock + 14 custom tools via MCP protocol, JSON-RPC 2.0) performing probabilistic root-cause analysis — fusing evidence from 8 independent verification sources using calibrated Bayesian scoring (Log-Likelihood Ratios). Achieved **54.8% time savings** and **97% accuracy** across 67K+ automotive data signals and 1,128 software modules. Adopted by 2+ engineering teams.
- Built a source-code dependency graph engine extracting 1.6M+ inter-module relationships from C source via pattern-based analysis across 3 repositories. Signal-level verification eliminates **75% of false positives** in failure attribution. Replaced manual export (35% incomplete) with a live git-based pipeline.
- **RAG Chatbot + KPI Dashboard:** Deployed hybrid BM25+ANN retrieval with cross-encoder reranking and LangGraph orchestration across 3 enterprise data sources (issue trackers + wiki). **77% answer rate**, **5.5× latency reduction** (22s → 4s) via semantic caching and async parallelism.
- **ASPICE SPOT Check (SWE.6):** Architected Agent-to-Agent (A2A) system using JSON-RPC 2.0 for automated automotive quality compliance assessment across **60+ evaluation dimensions** with mandatory source-citation at every step, hierarchical scoring, and multimodal ingestion (7 formats).

**Moshi Moshi** — Jun 2025 – Aug 2025
*AI Engineer Intern* — *Bengaluru, India*

- Built an automated lead generation pipeline using Selenium web scraping + open-source LLMs for prospect enrichment and personalized cold email generation via SMTP. Automated **∼50%** of the sales workflow end-to-end.
- Developed **Inspira** — a GenAI tool for designers to generate creative website concepts and content for branding clients. Reduced design ideation turnaround by **35%** and development handoff time by **25%**.

## Technical Skills

**ML/DL:** PyTorch, HuggingFace Transformers, LoRA/QLoRA/PEFT, XGBoost, scikit-learn, Bayesian Inference
**LLM Systems:** LangGraph, LangChain, RAG, MCP Protocol, A2A/ReAct Agents, Tool-Use, vLLM, Ollama
**Retrieval:** BM25, ANN, Cross-Encoders, Reciprocal Rank Fusion, ChromaDB, LanceDB, Sentence-Transformers
**Cloud & Infra:** AWS (Bedrock), Docker, Git, CI/CD, FastAPI, Streamlit, PySide6/Qt
**Engineering:** Python, asyncio, SQL, NumPy, Pandas, SciPy, NetworkX, Regex Engines, ProcessPool/ThreadPool

## Additional

**Teaching:** Teaching Assistant – Deep Learning & AI courses, IIT Patna
**Certification:** Deep Learning Specialization – DeepLearning.AI (Andrew Ng)
**Coursework:** Deep Learning, Natural Language Processing, Information Retrieval, Probabilistic Graphical Models